

Boiling pressure drop, local heat transfer distribution and critical heat flux in helical coils with R123

B.K. Hardik, S.V. Prabhu*

Department of Mechanical Engineering, Indian Institute of Technology, Bombay, India

ARTICLE INFO

Keywords:

Local heat transfer coefficient
Two-phase pressure drop
Critical heat flux
Flow boiling
Infra-red thermal camera

ABSTRACT

The objectives of the present work are to study the influence of curvature on local heat transfer coefficient, two-phase pressure drop and CHF in helical coils. The curvature of coil generates secondary flow of fluid which affects the mechanism inside helical coils. Very less information is available on the influence of curvature on flow boiling system. The present work investigates the characteristics of single phase flow and flow boiling in helical coils. Infra-red thermal imaging technique is used to measure the local wall temperature in the axial direction (along the length of coil) and in the circumferential direction. Experiments are performed with six helical coil test sections made of SS 304 circular tubes having three different tube diameters with two different coil diameters. The tube inner diameters are varying from 5.5 to 9.5 mm with wall thickness of 0.25 mm. The coil diameter to the tube diameter ratio ranges from 14 to 58 and coil pitch is 50 mm. The system pressure is varying from 1.1 bar to 3.6 bar. The analysis is performed for different heated lengths of test section. The experimental data of pressure drop and CHF are compared with available helical coil correlations. In fully developed flow boiling region, the circumferential averaged heat transfer coefficient in helical coils is same as in straight tubes. Correlations are suggested to predict local heat transfer coefficient in single phase and in flow boiling process, two-phase pressure drop and CHF in helical coils.

Nomenclature		
Symbol	Definition	Unit
Cp	Specific heat at constant pressure	J/kgK
d	Diameter of tube	m
D	Diameter of helical coil	
D/d	Diameter Ratio	
f	Friction factor	
G	Mass flux	kg/m ² s
g	Gravitational constant	m/s ²
h	Heat transfer coefficient	W/m ² K
I	Current	A
i	Enthalpy	J/kg
k	Thermal conductivity	W/m K
L	Length	m
$\dot{m}$	Mass flow rate	kg/s
P	Pressure	N/m ²
p	Pitch	m
Q	Heat supply	W

q''	Heat flux	W/m ²
T	Temperature	°C
V	Voltage	V
x	Quality of steam	
X, X _{LM}	Lockhart Martinelli parameter	
$X = \left(\frac{1-x}{x} \right)^{0.9} \left(\frac{\rho_g}{\rho_l} \right)^{0.5} \left(\frac{\mu_l}{\mu_g} \right)^{0.1}$		

Greek Definition

Ø	Two phase flow multiplier	$\Phi^2 = \frac{\Delta P_{TP,fric}}{\Delta P_{1P}}$
μ	Dynamic viscosity	N·s/m ²
ρ	Density	kg/m ³
σ	Surface tension	N/m
α	Void fraction	

Subscript Definition

acc	Acceleration
atm	Atmospheric
b	Bulk
conv	Convection
fg	Fluid to gas
fric	Friction

* Corresponding author. Department of Mechanical Engineering, Indian Institute of Technology, Bombay, Powai, Mumbai, India.
E-mail addresses: h.kothadia@gmail.com (B.K. Hardik), svprabhu@iitb.ac.in (S.V. Prabhu).

<i>g</i>	Vapour	
<i>h</i>	Heated	
<i>in</i>	Inlet	
<i>l</i>	Liquid	
<i>lo</i>	Liquid only	
<i>P</i>	Phase	
<i>rad</i>	Radiation	
<i>s</i>	Straight tube	
<i>c</i>	Helical coil	
<i>sat</i>	Saturated	
<i>SC</i>	Subcooled	
<i>sys</i>	System	
<i>w</i>	wall	
Abbreviation	Definition	
<i>CHF</i>	Critical Heat Flux	W/m^2
<i>HTC</i>	Heat Transfer Coefficient	W/m^2K
<i>R123</i>	2,2-Dichloro-1,1,1-trifluoroethane	
Dimensionless number		
<i>Bo</i>	Boiling number	$Bo = q''/G i_{fg}$
<i>Ja</i>	Jakob number	$Nu = h \cdot d/K$
<i>Nu</i>	Nusselt number	
<i>Pr</i>	Prandtl number	$Pr = \mu \cdot Cp/K$
<i>Re</i>	Reynolds number	$Re = 4\dot{m}/\pi d\mu$
<i>Dn</i>	Dean number	$Dn = Re\sqrt{d/D}$

1. Introduction

Helical coils are widely used in chemical, mechanical, biological, petrochemical and nuclear industries due to its compact structures, high heat transfer coefficients and good thermal expansion performances. Helical coiled steam generators (HCSG) are of major interest in the nuclear industry for electricity production [1]. Helically coiled steam generator is an attractive option for marine nuclear reactor. Study of boiling mechanism inside a helical coil tube is important in helical coil steam generator [2]. Flows in helical tubes are more complex than those in straight tubes due to secondary flows generated inside helical coils [3]. Pressure for intensification has increased the demand to improved understanding of fluid flow and heat transfer encountered in single phase flow and in flow boiling process in helical coils.

Hardik et al. [3] conducted a detailed review on available correlations for single phase heat transfer coefficient. Most of the correlations are validated for water and other high Prandtl number fluids. Very less information is available on the validation of heat transfer coefficient correlation with refrigerants. Local heat transfer correlations are suggested to calculate circumferential averaged heat transfer coefficient on inner side, outer side and total circumference of a helical coil with water as the working medium.

Naphone and Wongwises [4], Vashisth et al. [5], Fsadni and Whitty [6,7] and many others reviewed two-phase flow in helical coils. Reviews concluded that even though there is a wide application of helical coils in heat transfer industries, there is very less work available on flow boiling in helical coils compared to straight tubes. There are no general correlations available for boiling heat transfer coefficient and two-phase pressure drop. Owhadi et al. [8] carried out pioneering research on pressure drop and heat transfer coefficient in helical coil flow boiling system with water at atmospheric pressure. The work presented the circumferential variation in heat transfer coefficient in helical coil during flow boiling process. Most of the previous studies show overall averaged or circumferential averaged heat transfer coefficient in helical coils [2,9–13]. Recently, Santini et al. [1] presented circumferentially averaged and axially local heat transfer coefficient (HTC) in helical coil. The HTC is shown at twenty different axial location of a helical coil. To the authors' knowledge, literature showing variation in heat transfer

coefficient across the circumference of helical coil in flow boiling process is not described well. There is no work reported to compare the heat transfer on inner side and outer side of helical coil with straight tube for high density ratio fluids. Most helical coil heat exchangers used for industries and commercial application require the information about local heat transfer coefficient for design purposes to improve the effectiveness of heat exchanger. Many investigations have been carried out on experimental local heat transfer coefficient in single phase flows in helical coils which can be taken as the basis for investigation on two-phase flow in helical coils.

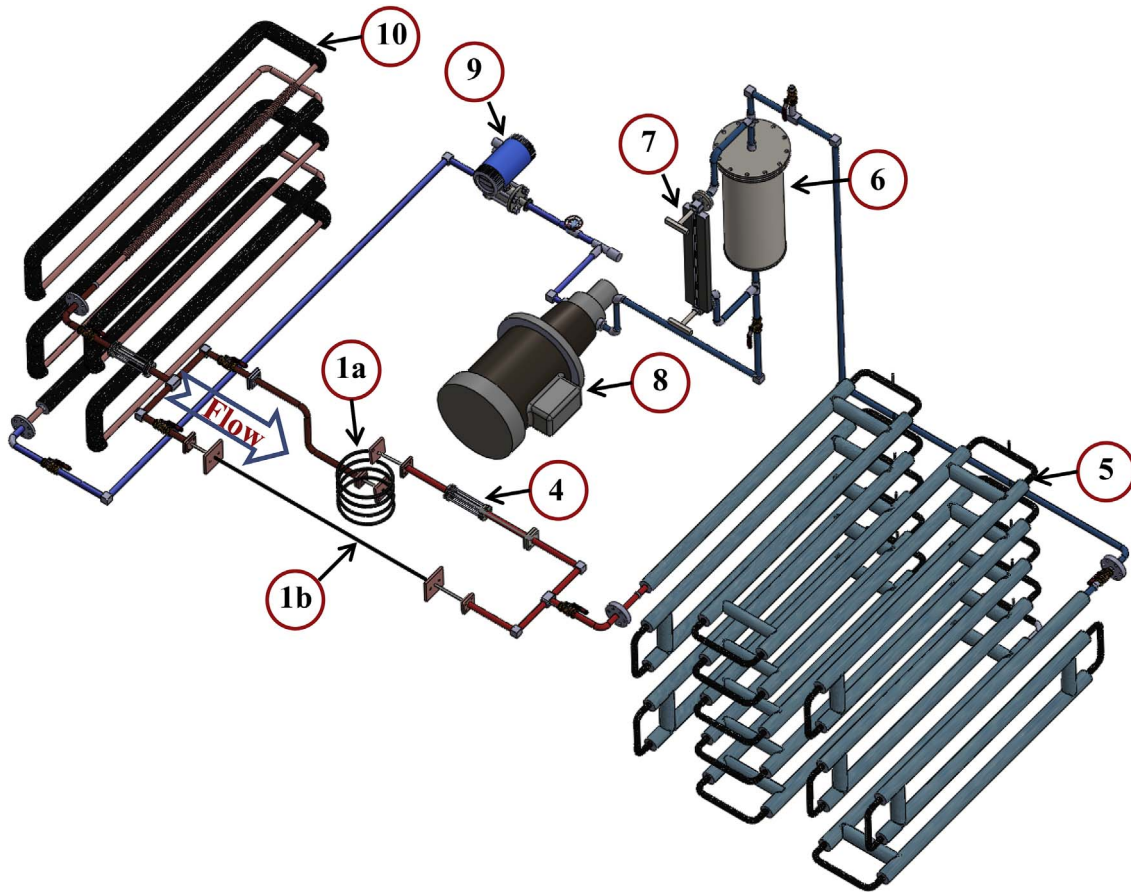
Hardik and Prabhu [14] conducted the study on local heat transfer coefficient and pressure drop in helical coils with water at low system pressure. They concluded that most of the literature available on two-phase pressure drop in helical coils published their own correlations for specific range of application [2,12,15–18]. This may be because of the complexity in helical coil due to secondary flow. They compared local heat transfer coefficient along the axial length of helical coils with different correlations. They concluded that the heat transfer coefficient in saturated flow boiling process is same as that in straight tubes. Kandlikar [19] correlation for saturated flow boiling heat transfer coefficient is working reasonably for helical coils. Heat transfer coefficient in subcooled boiling process is higher than straight tube. Two-phase pressure drop in helical coils is higher than straight tubes. Two-phase pressure drop in helical coil is accurately predicted using straight tube two-phase flow pressure drop multiplier and single phase pressure drop of helical coils. Hence, the effect of coil curvature in two-phase flow is similar to single phase flow.

Flow boiling heat transfer represents one of the most efficient types of heat transfer modes. Flow boiling plays an important role in the design and analysis of evaporators and condensers. However, it is restricted by the critical heat flux. The review works available on helical coil literature are limited to two-phase pressure drop and heat transfer coefficient. The work carried out by Owhadi et al. [8] concluded that helical coil should be considered in an application to convert liquid into dry steam. This conclusion is subsequently confirmed by later research work of Jayanti and Berthoud [20]. The study presented the total dryout data of Roumy et al. [20] experiments. The data show the total dryout at quality more than 0.9. Recently, Hardik and Prabhu [21] conducted the study on CHF in helical coils with water at low system pressure. They collected the literature available on CHF in helical coils. There is still lack of literature on the critical heat flux in helical coils. There is no general correlation available on CHF in helical coils. There is no data bank available for CHF in helical coil similar to straight tube CHF data bank.

The objectives of the present work are to study the local heat transfer coefficient, two-phase pressure drop and CHF in helical coils with R123 as the working fluid. The present work investigates the characteristics of single phase flow and flow boiling in helical coils. Infra-red thermal imaging technique is used to measure the local wall temperature in the axial direction (along the length of coil) and in the circumferential direction. Experiments are performed with six helical coil test sections made of SS 304 circular tubes having three different tube diameters with two different coil diameters. The tube inner diameters are varying from 5.5 to 9.5 mm with wall thickness of 0.25 mm. The coil diameter to the tube diameter ratio ranges from 14 to 58 and coil pitch is 50 mm. The system pressure is varying from 1.1 bar to 3.6 bar. The analysis is performed for different heated length of given test section.

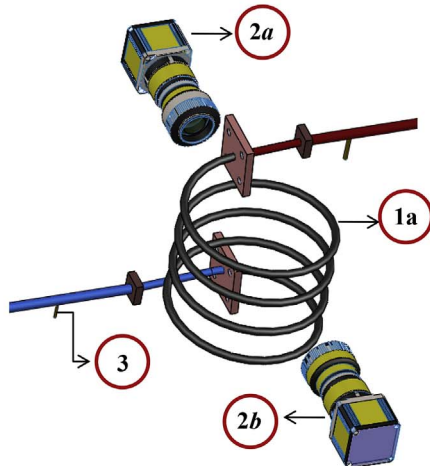
2. Experimental set-up

Fig. 1 presents a schematic diagram of the test facility constructed for this study. It is a closed loop system with refrigerant R123 serving as a working fluid. The test facility consists of an insulated fluid reservoir, magnetically coupled sealless gear pump, pre-heater, condenser, ball valves, straight horizontal tube test section and helical coil test section.

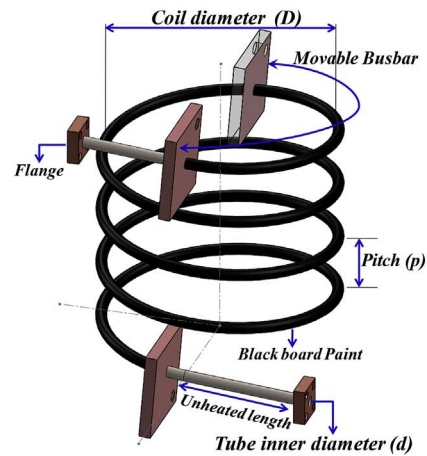


(A) Isometric view of experimental set-up

(1) Test section; a - Helical coil, b - Straight Tube, (2) IR Camera; a - 1st Position, b - 2nd Position, (3) Pressure Tap (4) Sight Glass (5) Condenser, (6) Reservoir, (7) Level Indicator, (8) Pump, (9) Flow Meter, (10) Pre-heater



(B) Location of thermal camera



(C) Helical coil test section

Fig. 1. Schematic of the experimental set-up.

The test facility is well insulated thermally from the atmosphere to minimize heat loss to the environment. Refrigerant R-123 has high boiling point temperature ($T_{\text{sat}} = 30.1^\circ\text{C}$ at $P = 1.1$ bar). Due to its high boiling point, water is used as secondary fluid to condense vapour

in condenser and repeat the cycle. Secondary water circuit is open loop cycle with uniform water inlet temperature. This makes system less complex, more stable and required less time to reach steady state. The gear pump with a mass flow rate range from 0 to 360 g/s is driven by a

Table 1
Geometrical and operating parameters of study.

Coil No	Tube Characteristics			Operating Characteristics					No. of CHF Data Points
	Inner diameter	Coil diameter	Heated length	System pressure	Density ratio	Mass flux	Heat flux	Quality	
	d (mm)	D(mm)	L _h (mm)	P _{sys} (bar)	ρ_l/ρ_g	G (kg/m ² s)	q" (kW/m ²)	x	
1	9.5	200	504, 610, 1006, 1524, 2146	1.5–3.6	151–62	145–1012	30–178	–0.2–1.06	31
2	7.5	150	467, 937, 1420	1.4–3.5	161–64	206–1623	30–237	–0.2–1.09	27
3	5.5	144	482, 970, 1472	1.3–3.6	171–62	255–1501	30–344	–0.2–1.17	17
4	9.5	300	506, 989, 1508	1.5–3.0	151–75	180–1058	30–201	–0.2–0.85	22
5	7.5	300	513, 1006, 1529	1.4–3.4	161–66	217–1195	30–296	–0.2–1.03	26
6	5.5	300	476, 984, 1472	1.1–3.6	207–62	392–1421	30–246	–0.2–0.92	17

D. C. motor. The speed of the D. C. motor is varied between 0 and 3500 rpm by a motor controller.

2.1. Measurement system

The temperature, pressure and flow rate measuring devices are installed at several locations to measure the pressure drop and heat transfer coefficient accurately in the test sections. Thermocouples (K-type) and pressure transmitters (Keller PA-21Y/10 bar) are installed at the inlet and outlet of the test section to measure the bulk fluid temperature and system pressure. Differential pressure transmitters are installed across the test section to measure the pressure drop across the test section. The range of the differential pressure transmitters are 0.622–62.2 kPa and 0–5 bar. Differential pressure transmitters are fixed on the same pressure taps of pressure transmitters. The locations of the pressure taps are shown in Fig. 1 (B). The taps are drilled 100 mm before the inlet of the test section and 50 mm after the exit of the test section in disturbance free part of the experimental set-up. Smoothness of inner surface of pipe is checked after drilling of vents and brazing of copper tube on pipe vent for pressure taps. Thermocouples are fixed on the surface of copper pipes at opposite diagonal end of pressure taps. Sight glasses are installed at the inlet and exit of the test section for a direct observation of the bubbles across the test section. This is recorded by means of a video camera.

Coriolis mass flow meter (EmersonCMF025 with Transmitter 1700R) is used to measure the mass flow rate of fluid. Flowmeter has capability to measure the density of fluid and bulk fluid temperature. Infra-Red thermal camera (Make: Themoteknix, Model: VisIR640) is used to measure the test section local wall temperature without disturbing the test sections. Thermal camera measures the intensity of infra-red radiation emitted by the test section wall. The intensity of radiation depends on the emissivity and temperature of wall surface. The test section is painted with a thin coat of high temperature black board paint to have a uniform emissivity of 0.85 for infra-red radiation. The thermal camera is set at two different positions as shown in Fig. 1 (B) to cover the complete helical coil.

DC power supply (Make: Aplab, Model: CVCC50kW; 0–40 V voltage and 0–1250 amp current) is used to supply power through the test section. The copper electrodes are connected to the test section and a DC power supply system via copper cables. All the test sections are heated electrically by passing DC current through the tube wall maintaining the uniform heat flux condition. The heat supplied is determined from the current and voltage across the test section. The current is measured using the power supply digital reading and the voltage is measured using multimeter.

The inlet temperature of the refrigerant is controlled and maintained at the desired temperature by a preheater and condenser. Preheater (2000 W) is heated by passing AC current through a Nichrome wire mounted on the wall of pre-heater. Heat supplied in the pre-heater and the test section is removed in the condenser. A counterflow tube-in-tube heat exchanger located at the outlet of the test

section condenses the evaporated refrigerant. Then, the refrigerant goes back to the reservoir, thus closing the loop. Electromagnetic mass flow meter (Make: Manas, Model: SR1000P-DN25-PTFE) and K-type thermocouples are installed in the condenser to measure the heat absorbed by the water. All the thermocouples are inserted into the fluid to measure the actual water temperature at the inlet and outlet of the condenser.

The test facility is controlled and monitored using an experimental control panel and a personal computer using a data acquisition system (DAS) (Agilent 34972A), respectively. All the thermocouples are connected directly to DAS. The flowmeters and pressure transmitters are connected to DAS through 100 ohms resistors to convert current signal (4–20 mA) into readable voltage signal (0.4–2 V). All the readings are taken after the complete system reaches steady state.

Fig. 1 (C) shows an exploded view of the test section. Helical coils used for testing are made of stainless steel SS304. The distance between the upper and bottom electrode is equal to the heated length of the test section. There is a non-heated coil length of approximately minimum 150 mm at each end of the tube to stabilize the fluid flow and avoid the flow entrance effect. The moveable bus bar is adjusted at different heated lengths to study the effect of length on CHF. The geometric details of the six coils tested are given in Table 1. Three different tube diameters with two different coil diameters are used for analysis of pressure drop and heat transfer coefficient. Number of critical heat flux data collected for specific test section is given in Table 1.

2.2. Operating procedure

Steady-state characteristics of the system are first obtained on the pressure drop versus mass-flux plane. The experiment is conducted by fixing the mass flux at a desired value with a pre-selected system pressure, while increasing the heat flux to the test section. Then the flow rate of water in condenser is adjusted and preheater (if required) turned on so that a desired inlet subcooling condition reached at the inlet of the test section. Next, the heat flux to the test section is increased to a starting value. Various data like pressures, temperatures, mass flow rate and pressure drop are recorded simultaneously after thermal equilibrium is attained in the system. Thermal camera captured ten thermal images of test section from both sides two times to get better accuracy in the wall temperature measurement. Thereafter, heat flux is increased by a small increment repeatedly until the next desired data point is reached. The wall temperature is observed using the display on the display of thermal camera, while the heat flux is increasing. All the required data are recorded and cycle is repeated until the wall temperature becomes too high. The above procedure is repeated by changing other parameters alternatively.

For each test section, the test matrix covered a wide range of operating conditions to investigate the effect of the different parameters on local heat transfer coefficient, two-phase pressure drop and CHF. The variables used in analysis of the present study are mass flux, heat flux, coil diameter and tube diameter. The operating parameters

covered in the present study are listed in Table 1.

3. Data reduction

Procedure to measure the inner surface wall temperature is given in this section. Bulk fluid temperature calculated in single phase flow and in flow boiling process is discussed. Equations required to calculate heat transfer coefficient and pressure drop are presented. Methods used to predict CHF are explained. The uncertainties in the measurement of different parameters are presented.

3.1. Temperature measurement

The bulk fluid temperature at inlet and exit of the helical coil is measured using thermocouples. During two-phase flow condition, thermocouple at the exit of helical coil shows the temperature same as saturation temperature at that pressure measured using pressure transmitter. The local bulk fluid temperature in subcooled boiling region is measured by the linear interpolation between inlet temperature and saturation temperature. The saturation temperature at zero quality (at $x = 0$) is based on inlet system pressure. The local saturation temperature in saturated boiling region is calculated from local pressure. Interpolation pressure profile between inlet and exit pressure is calculated from pressure drop profile. To draw pressure drop profile accurate, the value of constant is changed in the pressure drop correlation Eq. (17) such that pressure drop from the correlation is equal to experimental pressure drop.

Thermo-physical properties of liquid in the single phase flow and in the subcooled flow boiling calculation are calculated at local bulk fluid temperature. In the saturated flow boiling calculation, the properties are calculated at the local saturation temperature. The vapour properties in all the calculations are calculated at local saturation temperature. All the thermo-physical properties used in the calculations are calculated from National Institute of Standards and Technology (NIST) tabulated values.

Local wall temperature distribution of the helical coil is obtained by averaging ten thermal images for each configuration. Fig. 2 shows the cross section of helical coil with illustrative direction of secondary flow. Local points show the circumferential temperature points of thermal image. Circumference from 270° to 0°-90° is considered as outer side, 90°-180°-270° is considered as inner side and complete circumference as total. Thermal images measure the outer surface wall temperature of

the helical coil. The wall thickness of all the test sections is 0.25 mm. Hence, Inner surface wall temperature is calculated from measured outer surface wall temperature using one dimensional, steady state heat conduction method with uniform volumetric heat generation across the helical coil wall due to Joule heating.

3.2. Heat transfer coefficient

The heat supplied to the test section is calculated from Eq. (1), where V and I are voltage across the test section and supplied electric current. Heat loss from the helical coil outer wall surface to the atmosphere for a given heat load is subtracted from the total heat supply. Convective and radiation heat loss are calculated theoretically. However, test section heat loss calibration tests are performed in single phase flow to validate the calculation method. The procedure to calculate heat loss is given in Hardik and Prabhu [14]. In the present study, the percentage of heat loss is varying from 0.2% to 6% of total heat supply.

$$q'' = \frac{Q_{conv}}{\pi d L_h} = \frac{Q - Q_{loss}}{\pi d L_h} = \frac{VI - (Q_{conv,atm} + Q_{rad,atm})}{\pi d L_h} \quad (1)$$

The local thermodynamic equilibrium quality and local heat transfer coefficient in subcooled and saturated boiling regions are calculated from Eqs. (2)–(4). An equation (2) gives the quality at specific length of test section. The length of the test section from where, the subcooled flow boiling transits into saturated flow boiling (zero quality) is calculated from Eq. (2) by keeping $x_{local} = 0$. In a subcooled region x_{local} gives negative values while in saturated region x_{local} gives the positive values.

$$x_{local} = \frac{\frac{Q_{conv} L_{local}}{\dot{m} L_h} - i_{in}}{i_{fg}} \quad (2)$$

$$i_{in} = C_p (T_{sat} - T_{in}) \quad (3)$$

$$h_{local} = \frac{q''}{T_{w, local} - T_{b, local}} \quad (4)$$

Local heat transfer coefficient in a single phase turbulent flow for straight tube is calculated using Dittus-Boelter correlation as given in Eq. (5). Single phase local heat transfer coefficient on inner side, outer side and total in helical coil is calculated from Hardik et al. [3] correlations as given in Eqs. (6)–(8).

$$h_{l-local} = 0.023 Re_{local}^{0.8} Pr_{local}^{0.4} \frac{k_l}{d} \quad (5)$$

Location	Correlation
Total	$h_{l-Total} = 0.0456 \left(\frac{D}{d}\right)^{-0.16} Re_{local}^{0.8} Pr_{local}^{0.4} \frac{k_l}{d}$

(6)

Location	Correlation
Outer Side	$h_{l-Outer} = 0.104 \left(\frac{D}{d}\right)^{-0.315} Re_{local}^{0.8} Pr_{local}^{0.4} \frac{k_l}{d}$

(7)

Location	Correlation
Inner Side	$h_{l-Inner} = 0.012 \left(\frac{D}{d}\right)^{0.088} Re_{local}^{0.8} Pr_{local}^{0.4} \frac{k_l}{d}$

(8)

3.3. Pressure drop

The experimental pressure drop measured across the pressure taps in a helical coil is the sum of pressure drop in single phase flow, subcooled boiling flow, saturated boiling flow and adiabatic two-phase flow. Fig. 3 shows illustrative diagram of the total pressure drop summed with pressure drop in different flow regions.

$$\Delta P_{Total} = \Delta P_l + \Delta P_{sc} + \Delta P_{TP} \quad (9)$$

Pressure drop in single phase flow in a straight tube is calculated

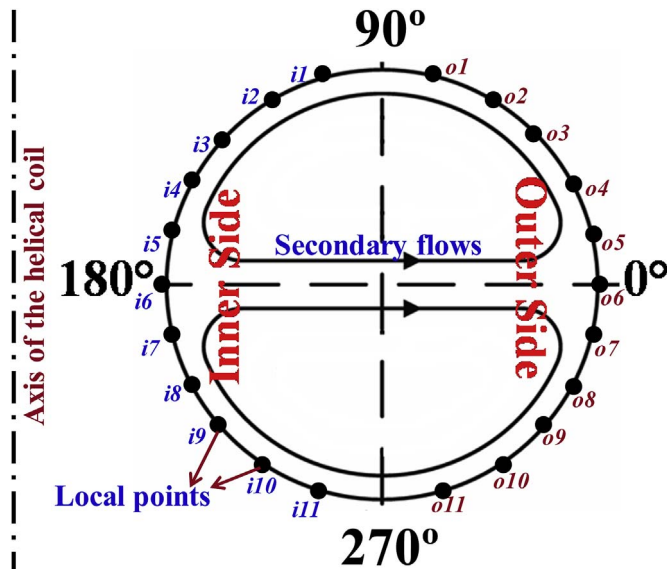


Fig. 2. Indication of local points for wall temperature measurement on the circumference of a helical coil.

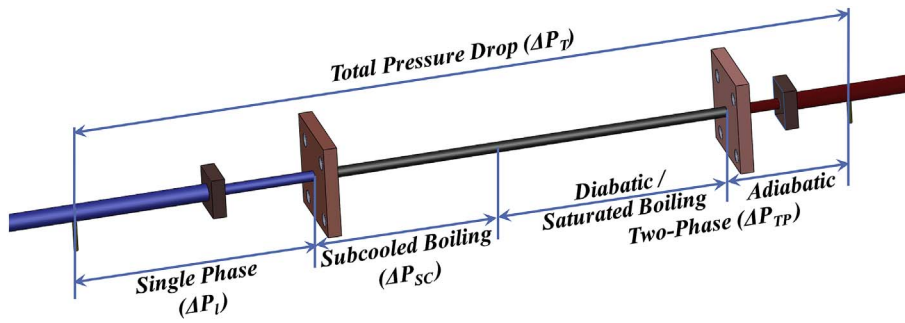


Fig. 3. Different regions of pressure drop.

using Blasius friction factor correlation (Eq. (10)) and in a helical coil is calculated using Ito's correlation (Eq. (11)) as suggested by Hardik et al. [3]. The subcooled flow boiling pressure drop is calculated using Baburajan et al. [22] correlation as given in Eq. (12). The tube diameter (d) is in mm and ΔP_{1P} is adiabatic single phase pressure drop with properties taken at saturated temperature. Single phase pressure drop in Eq. (12) is calculated using Ito's correlation instead of straight tube correlation which is used by Baburajan et al. [22] for straight tube. Single phase pressure drop and subcooled boiling pressure drop are much lower than two-phase pressure drop.

$$f_s = 0.079Re^{-0.25} \quad (10)$$

$$\Delta P_{1P} = f_c \frac{32}{\pi^2} \frac{\dot{m}^2 L}{\rho d^5} ; f_c = f_s (Re(d/D)^2)^{\frac{1}{20}} \quad (11)$$

$$\frac{\Delta P_{sc}}{\Delta P_{1P}} = 1 + 32500Bo^{1.6}Ja^{-1.2} \left(\frac{d}{9.53} \right) \quad (12)$$

The two-phase pressure drop in a flow boiling system is the sum of frictional pressure drop, pressure drop due to fluid acceleration and gravity as in Eq. (13). The acceleration pressure drop is calculated from separated flow model and void fraction is calculated from Steiner equation (Didi et al. [23]) as shown in Eq. 14 and 15;

$$\Delta P_{TP} = \Delta P_{fric} + \Delta P_{acc} + \Delta P_{gravity} \quad (13)$$

$$\Delta P_{acc} = G^2 \left\{ \left[\frac{(1-x)^2}{\rho_l(1-\alpha)} + \frac{x^2}{\rho_g \alpha} \right]_{out} - \left[\frac{(1-x)^2}{\rho_l(1-\alpha)} + \frac{x^2}{\rho_g \alpha} \right]_{in} \right\} \quad (14)$$

$$\alpha = \frac{x}{\rho_g} \left[(1 + 0.12(1-x)) \left(\frac{x}{\rho_g} + \frac{1-x}{\rho_l} \right) + \frac{1.18(1-x)[g\sigma(\rho_l - \rho_g)]^{0.25}}{G^2 \rho_l^{0.5}} \right]^{-1} \quad (15)$$

3.4. CHF identification method

Santini et al. [24] listed four definitions to identify the CHF in the test section. In the present study non-uniformity in the curve of heat flux variation with the wall temperature method is used for CHF identification. The wall temperature is measured using thermal camera without disturbing test section. During CHF experiments, heat flux is increased stepwise. After small increment in heat flux, the wall temperature from the thermal camera is captured. When the wall temperature of test section suddenly increases compared to the previous wall temperature by at least 50 °C, it is determined as CHF. CHF data are measured at dryout condition. In most of the cases, the dryout occurs at the exit of test section. In some cases, the dryout occurs near the exit of the test section. Hence, the thermodynamic equilibrium quality at the exit of the test section is nearly same as the critical quality.

3.5. Uncertainties in the measured and computed parameters

The uncertainties in different parameters are calculated based on the method suggested by Moffat [25] and Kline and McClintock [26]. Total uncertainty in parameter is calculated by considering the error in repeatability, reproducibility and uncertainty of the calibration system. The uncertainties in the helical coil tube diameter, coil diameter and length are obtained by measuring the values at different locations. Ovalization of tubes are measured after fabrication of helical coil test sections. Difference in major and minor diameter of tube due to ovalization is less than 2%. The uncertainties in mass flow rate, temperature, and pressure are calculated by comparing the reading of DAS with the value obtained from standard instruments. The uncertainties in heat flux resulted from the accuracy of the ammeter and voltmeter. The uncertainty of different parameters used in the experimental analysis is shown in Table 2.

4. Results and discussions

Infra-red thermal imaging technique for the measurement of wall surface temperature using thermal camera is validated and well developed by authors in their earlier works. The accuracy and reliability of the experimental system with refrigerant R123 are validated using a pressure drop and heat transfer experiments on straight horizontal tube before performing the experiments with helical coils. Single phase flow and flow boiling experiments are performed with the experimental setup on a straight horizontal tube and compared with well-developed correlations. Experimental single phase pressure drop in a straight tube is compared with the Blasius correlation for a Reynolds number range of 3000–30000. Local heat transfer coefficient of single phase flow in a straight tube is compared with Dittus-Boelter correlation. Two-phase pressure drop and local heat transfer coefficient in a straight tube with R123 are compared with Hardik and Prabhu [27] and Kandlikar [19] correlations, respectively. In the present work, the experiments are performed for single phase heat transfer coefficient in helical coils. Circumferential averaged heat transfer coefficient on inner side, outer side and total are compared with the correlations along the length of coil. Temperature and heat transfer distribution during the flow boiling

Table 2
Experimental uncertainties.

Parameter	Relative uncertainty (%)	Parameter	Relative uncertainty (%)
Pipe diameter	1	Pressure	1.5
Pipe length	0.5	Quality	3.4
Coil diameter	1	Heat flux	3
Current	0.5	Mass flux	2.7
Voltage	1	Boiling number	4
Bulk temperature	1.5	Heat transfer coefficient	11.4
Wall temperature	1.3	Two-phase pressure drop	9.9
Mass flow rate	1.8		

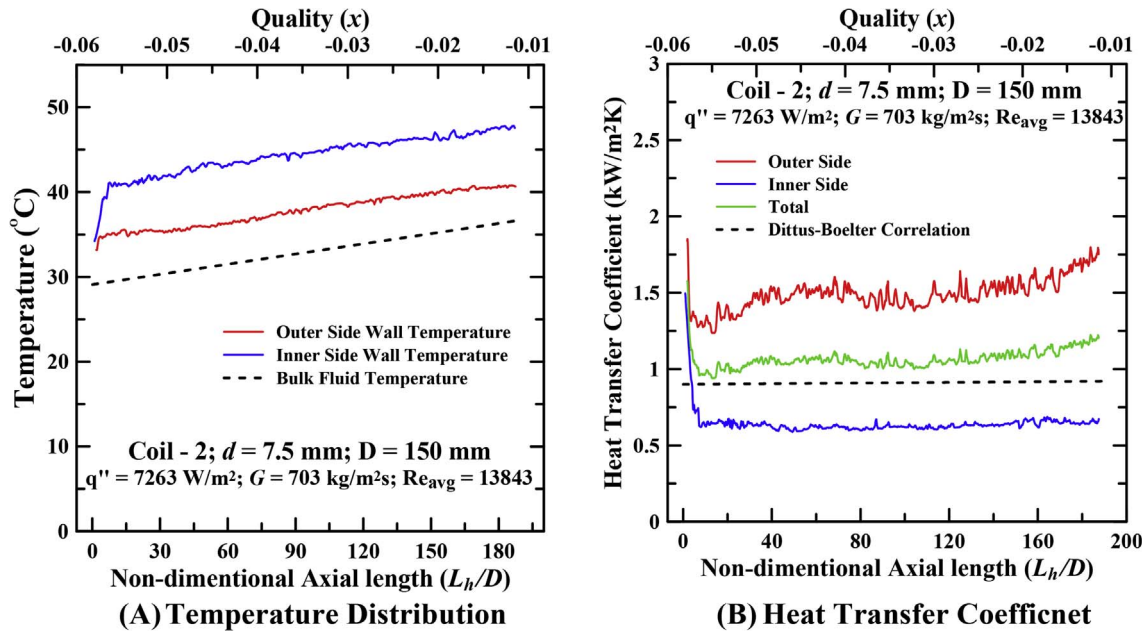


Fig. 4. Temperature and heat transfer coefficient distribution for Coil - 2.

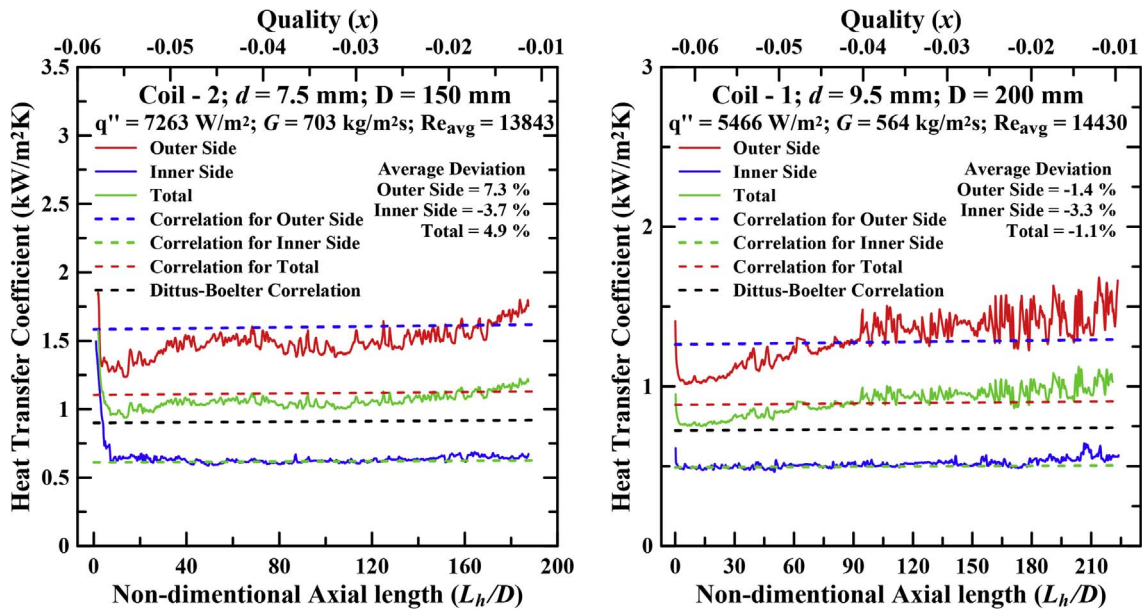


Fig. 5. Local HTC comparison with Hardik et al. [3] correlations.

process inside helical coils are analysed. Local heat transfer coefficient along the axial length of coil and two-phase pressure drop are compared with the available correlations. CHF data for different mass flux, tube diameter, coil diameter and coil length are collected and compared with the available helical coil correlations and straight horizontal tube correlations.

4.1. Temperature and heat transfer distribution in single phase flow

The wall temperature is measured on the circumference of a helical coil at different locations as shown in Fig. 2 and averaged for half circumference as inner side and outer side. Single phase temperature and heat transfer distribution for helical coil 2 with a tube diameter 7.5mm and D/d ratio 20 is shown in Fig. 4. Fig. 4 (A) shows the inner side wall temperature, outer side wall temperature and bulk fluid temperature distribution inside a helical coil. The outer side wall

temperature is lower than that on the inner side. This is due to the secondary flow direction of helical coil as shown in Fig. 2. In curved tubes, high density cold fluid moves on the outer side and low density hot fluid moves to the inner side due to the centrifugal force. The heat transfer coefficient is computed from measured temperatures. Local heat transfer coefficient along the length of helical coil is shown in Fig. 4 (B). Heat transfer coefficient on the inner side of a helical coil is much lower than that on the outer side. The ratio of outer side to inner side heat transfer coefficient is 2–3. Total circumferential averaged heat transfer coefficient is compared with the Dittus-Boelter correlation. Dittus-Boelter correlation is calculated at the same local properties using bulk fluid temperature. The total circumferential averaged heat transfer coefficient in a helical coil is higher than that of a straight tube. The enhancement in heat transfer decreases with the increase in D/d ratio and hence, with decrease in the strength of secondary flow and centrifugal force. The inner side heat transfer coefficient is lower than

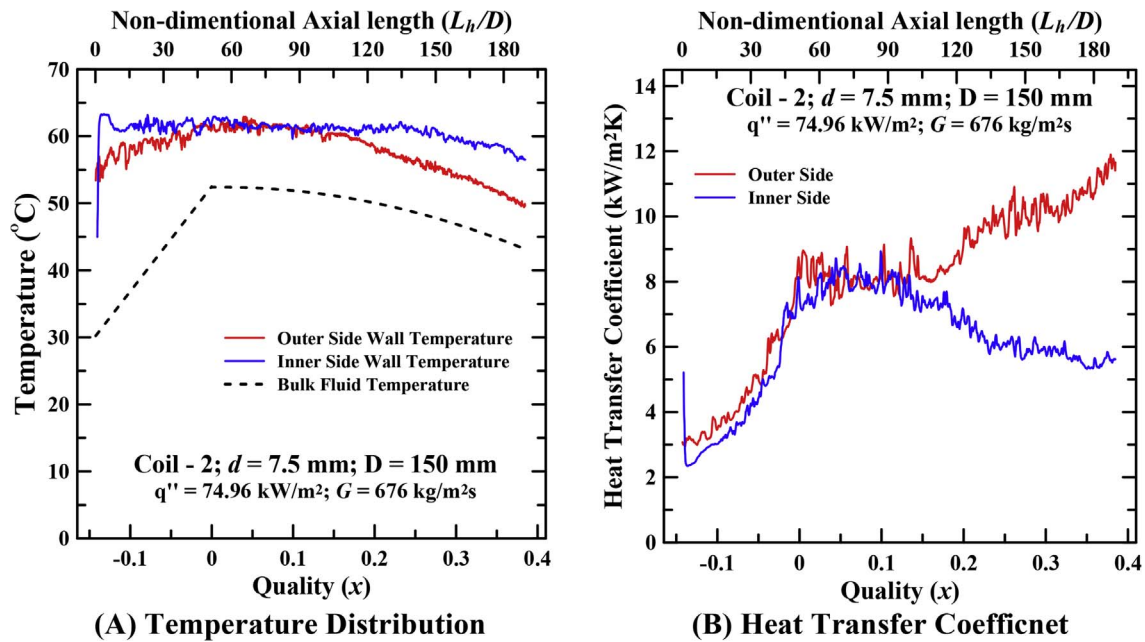


Fig. 6. Temperature and heat transfer coefficient distributions for Coil - 2.

Table 3
Comparison of experimental results with correlations reported in the literature.

Sr. No.	Author	Absolute Mean	Average	± 30% Range
1	Kandlikar [19] (1991) (Saturated)	11.1	−2.0	96.3
2	Gungor and Winterton [31] (1986) (Saturated)	20.8	13.1	77.8
3	Lazarek and Black [30] (1982)	21.9	−1.8	72.1
4	Shah [28] (1982)	18.5	−17.4	84.3
5	Schrock and Grossman [32] (1962)	25.8	−25.4	63.9
6	Liu and Winterton [29] (1991) (Saturated)	24.7	−23.4	57.8
7	Kandlikar [19] (1991) (Subcooled)	10.9	−3.2	100
8	Gungor and Winterton [31] (1986) (Subcooled)	15.6	−14.7	97.0
9	Liu and Winterton [29] (1991) (Subcooled)	34.1	−34.1	22.4
10	Shah [33] (1977)	17.0	−16.7	88.1
11	Moles and Shaw [34] (1972) (Subcooled) ($x < -0.03$)	121	121	0

$$\text{Absolute Mean Deviation} = \frac{1}{N} \sum_{i=1}^N [(|X_{cal} - X_{exp}|)/X_{exp}]$$

$$\text{Average Deviation} = \frac{1}{N} \sum_{i=1}^N [(X_{cal} - X_{exp})/X_{exp}]$$

that of straight tube while outer side is high than that of straight tube. As the D/d ratio increases, the inner side heat transfer coefficient increases and outer side decreases and hence, reaching towards the straight tube.

4.2. Comparison of heat transfer coefficient with correlation in single phase flow

Circumferential averaged inner side, outer side and total heat transfer coefficient computed experimentally are compared with the correlations Eqs. (6)–(8) respectively. Total circumferential averaged heat transfer coefficient is also compared with the Dittus-Boelter correlation to calculate the enhancement in a helical coil in comparison with the straight tube. The correlations derived for local heat transfer coefficient in helical coils are matching well with experimental data. Total 30 data points of experimental heat transfer coefficient in inner side, outer side and total circumferential averaged of all six test sections are compared with correlations. All the data points are matching with maximum 10% of averaged deviation for total, 8% for inner side and 13% for outer side. Sample comparison for helical coil 1 and 2 with tube inner dia. 9.5 mm and 7.5 mm and D/d ratio 21 and 20, respectively is shown in Fig. 5. Fig. 5 shows the enhancement in total average HTC of helical coil compared to straight tube. The enhancement in coil 1 is 19% and in coil 2 is 15%.

4.3. Temperature and heat transfer distribution in flow boiling process

The distribution of temperature and HTC during flow boiling is presented in this section. The fluid enters in subcooled condition to the helical coil and exits as a saturated mixture of liquid and vapour with some vapour quality. The distribution of inner side wall temperature, outer side wall temperature and bulk fluid temperature in subcooled and saturated boiling regions is shown in Fig. 6 (A). The temperature distribution in single phase flow inside a helical coil is discussed above. During single phase flow, inner side has high temperature. Hence, boiling starts first from inner side. The wall temperature on the inner side remains uniform from the inlet of the section. The wall temperature on the outer side increases till zero quality. This is due to the increase in bulk fluid temperature. The bulk fluid temperature distribution in a helical coil is not uniform. The low temperature fluid moves towards the outer side, while the high temperature fluid moves towards the inner side. Hence, on the inner side, boiling gets started and inner side wall temperature becomes uniform due to nucleate boiling. However, outer side wall temperature increases with the increase in the fluid temperature near to the outer surface. In nucleate boiling region,

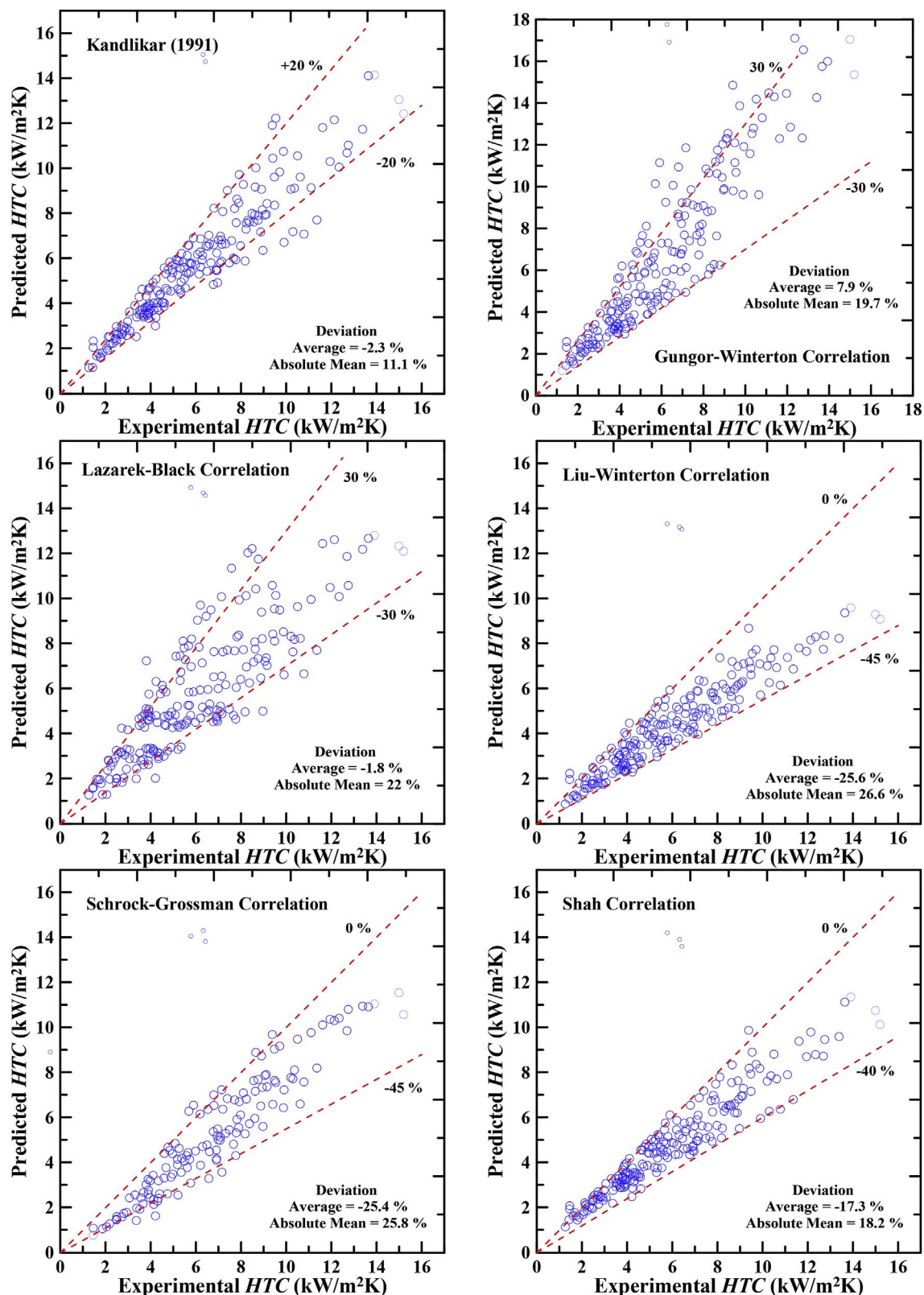


Fig. 7. Comparison of experimental heat transfer coefficient with available correlations.

the wall temperature on the inner side and the outer side remains almost same. In convective boiling region, wall temperature on the outer side decreases at a fast rate compared to the inner side. This is again due to the secondary flow movement in the annular liquid film. Bulk fluid temperature in subcooled and saturated region is calculated with

the method explained in data reduction section.

The heat transfer coefficient is calculated from measured wall and bulk fluid temperatures. The heat transfer distribution on inner side and outer side of helical coil during flow boiling process is shown in Fig. 6 (B). Heat transfer coefficient at the outer side is higher than the inner

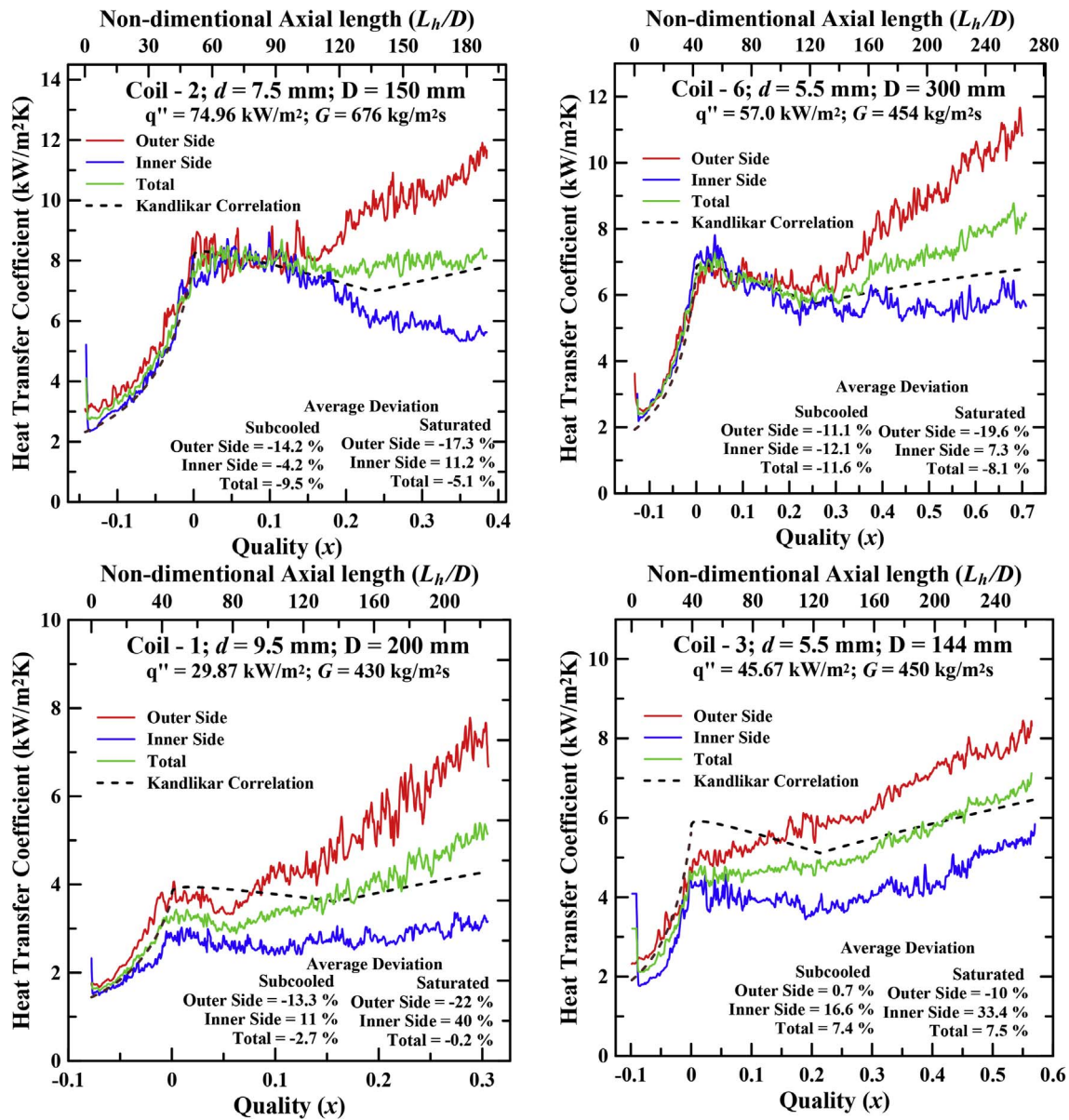


Fig. 8. Comparison of local HTC with Kandlikar [19] correlation.

side. In a subcooled boiling process, difference in the heat transfer coefficient between inner side and outer side decreases. It remains same during the nucleate boiling process and increases again during the convective boiling process. The outer side heat transfer coefficient increases in the subcooled boiling region, remains uniform in the nucleate boiling region and increases in the convective boiling region. The inner side of the helical coil has nucleate boiling process over the complete length of the coil. Hence, the inner side wall temperature remains almost uniform along the axial length of helical coil in complete boiling process. Heat transfer coefficient in the subcooled boiling region increases due to the increase in the bulk fluid temperature and then decreases due to the decrease in system pressure due to pressure drop.

4.4. Comparison of experimental heat transfer coefficient with the available correlations

The local heat transfer coefficient along the axial length of helical coil is measured for different test sections with different heat fluxes and mass fluxes. Total 200 data sets of local HTC are collected. These local HTC are averaged for heated length of coils and total 200 overall

averaged data points are collected. The data points include the results of pure subcooled flow boiling and mix (Subcooled + Saturated) flow boiling. These overall averaged data points are compared with available helical coil HTC correlations and straight tube correlations. Subcooled HTC results are compared with subcooled correlations and saturated HTC results are compared with saturated correlations. The analysis of comparison is given in Table 3 to identify the correlation works well to predict HTC in helical coils.

The comparison with selected correlations is shown in Fig. 7 to find the trend of results. Shah [28] correlation underpredicts the experimental data. The correlation is deviating from 0 to -40% with all the data points with mean deviation 18.2%. Liu and Winterton [29] correlation deviates by 0 to -45% with the experimental data with a mean deviation of 26.6%. The data are in centred with Lazarek and Black [30] and Gungor and Winterton [31] correlations but the scatter in prediction are higher. Schrock and Grossman [32] correlation underpredicts the experimental data. Kandlikar [19] correlation is working well with the present experimental data. The experimental data of helical coils are deviating within $\pm 20\%$ with the correlation. Hardik and Prabhu [14] compared the experimental data of helical coils for water

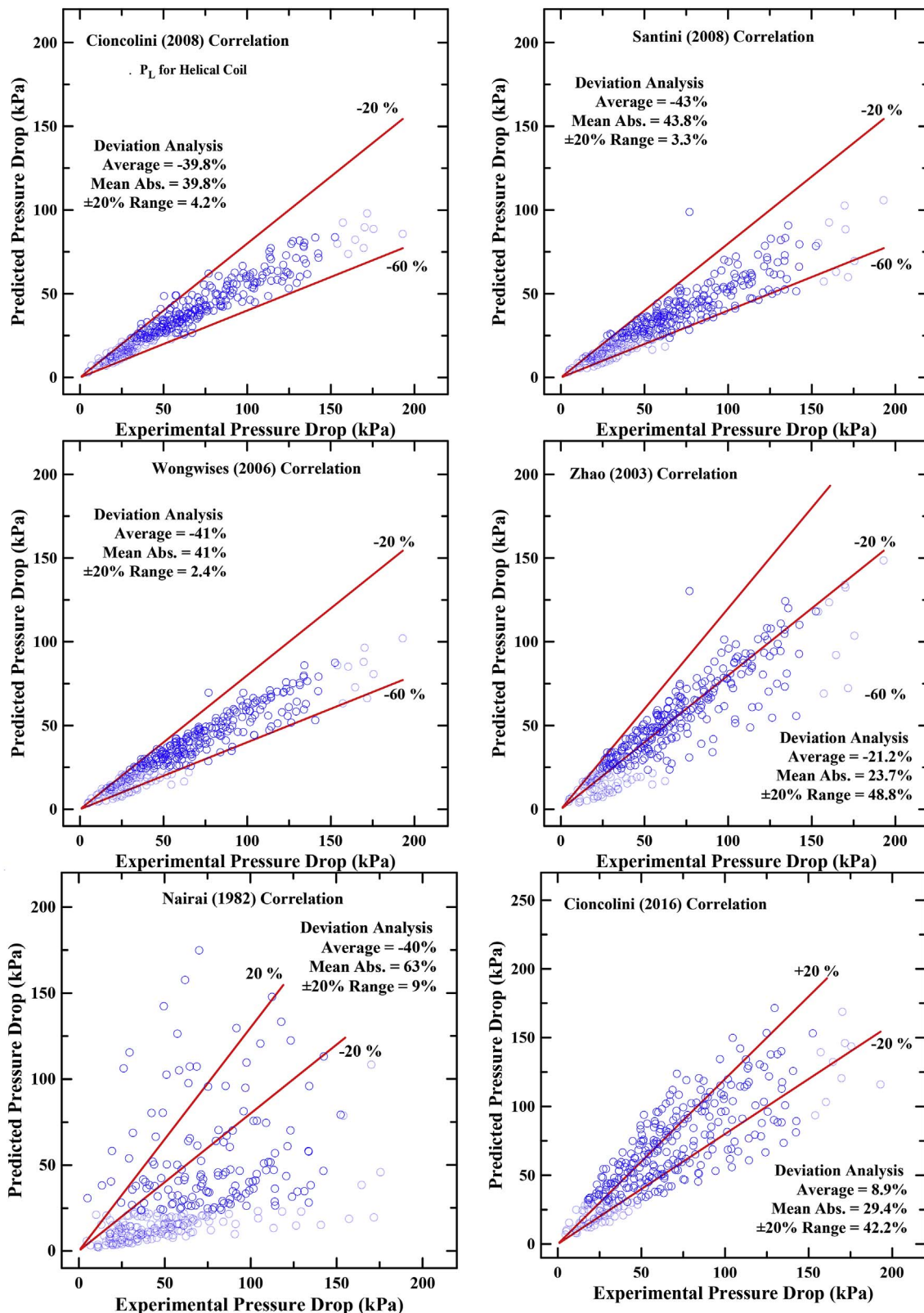


Fig. 9. Comparison of two-phase pressure drop data with helical coil correlations.

with the available HTC correlations. Comparison shows that the Kandlikar [19] correlation works well to predict the data points. The correlation predicts the present data with a mean deviation of 11.1%. Hence, overall Kandlikar [19] correlation works well for boiling heat transfer coefficient measurement.

4.5. Comparison of local heat transfer coefficient with Kandlikar correlation

Kandlikar [19] correlation works well for overall averaged heat transfer coefficient. In the present section, Kandlikar correlation is

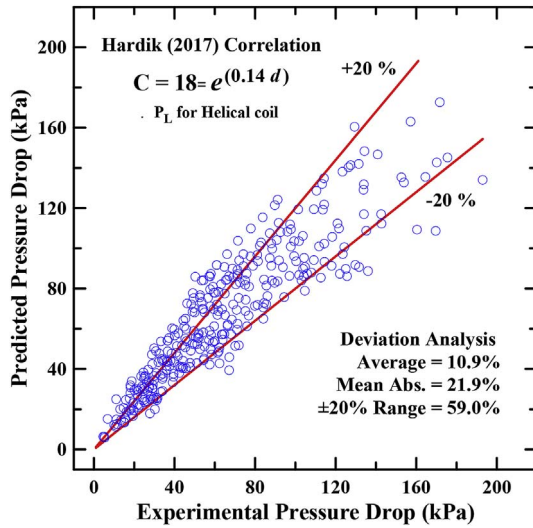


Fig. 10. Comparison of two-phase pressure drop data with Hardik and Prabhu [14] correlation.

Table 4
Deviation analysis of correlations from the experimental pressure drop.

Sr. No.	Author	Average	Absolute Mean	± 20%
1	Hardik and Prabhu [14] (2017)	10.9	21.9	59.0
2	Zhao et al. [12] (2003)	−21.6	23.8	48.8
3	Cioncolini and Santini [35] (2016)	8.9	29.4	42.1
4	Elsayed et al. [10] (2012)	−84.7	84.7	0
5	Cioncolini et al. [15] (2008)	−39.8	39.8	4.2
6	Santini et al. [17] (2008)	−43	43	3.3
7	Wongwises and Polsongkram [13] (2006)	−41.1	41.1	2.4
8	Guo et al. [16] (2001)	−77.8	77.8	0
9	Nairai et al. [2] (1982)	−40.2	62.8	9.0
10	Unal et al. [18] (1981)	89.9	89.9	0

compared with the circumferentially averaged HTC along the length of a helical coil. Circumferentially averaged heat transfer coefficient on inner side, outer side and total are compared with the correlation for different helical coils as shown in Fig. 8. The subcooled HTC is compared with the subcooled equation and the saturated HTC is compared with the saturated equation. Circumferential HTC variation in the subcooled region is lower. The correlation is matching well with circumferentially averaged HTC in subcooled region. In the saturated region, the inner side heat transfer coefficient is lower than the straight tube, while the outer side heat transfer coefficient is higher than that of the straight tube. Hence, the circumferential averaged total HTC is nearly equal to that of the straight tube. The correlation is changing the slope in the nucleate and convective boiling region. The correlation is matching with a reasonable accuracy with the circumferentially averaged HTC along the length of a coil in the saturated boiling region.

4.6. Comparison of two-phase pressure drop with available correlations

The pressure drop across the test sections are measured along with HTC and CHF. Total 332 data points of pressure drop are collected for saturated flow boiling process for different test sections. The data are compared with the available correlations of two-phase pressure drop in helical coils to identify an appropriate correlation which works well to predict the boiling pressure drop of helical coils. The comparison is shown in Fig. 9. Most of the correlations underpredict the present data points. Readings are consistent with Cioncolini et al. [15] correlation with a deviation of −20% to −60%. Santini et al. [17] and Wongwises

and Polsongkram [13] correlations are underpredicting the experimental results with a deviation of 20% to −60%. Zhao et al. [12] correlation slightly underpredicts the experimental results. The mean deviation with Zhao et al. [12] is 23.7%. Nairai et al. [2] correlation is not working well for the present experimental data. The scatter is higher with Nairai et al. [2] correlation. The correlation has a mean deviation of 62.8%.

Recently, Cioncolini and Santini [35] developed a general correlation for two-phase pressure drop in helical coils. This correlation is developed using the data set of Zhao et al. [12] and Santini et al. [17] of helical coils and data from three literature for straight tube. Data set includes both adiabatic and diabatic data. Hence, the correlation is applicable for helical coils and straight tubes. The correlation is validated for adiabatic and diabatic conditions. The correlation is in the form of a two-phase friction factor with density and viscosity being calculated by taking mixture average. The present data is compared with the Cioncolini and Santini [35] correlation as shown in Fig. 9. Correlation is matching reasonably well with the experimental data. Data are well centred with the correlation. The scatter is higher in the prediction with a mean deviation of 29.4%.

4.7. Comparison of two-phase pressure drop with Hardik and Prabhu correlation

Hardik and Prabhu [27] derived a correlation for the two-phase pressure drop in straight horizontal tubes. The correlation is used for two-phase pressure drop in helical coils by Hardik and Prabhu [14]. They concluded that the correlation works well for helical coils using single phase friction factor for helical coils. Present experimental data are compared with Hardik and Prabhu [14] correlation as shown in Fig. 10. The correlation is working well with present data. The mean deviation with correlation is 21.9%.

Deviation analysis of experimental data with all the available correlations is given in Table 4. Details of all the correlations used in Table 4 are given in Hardik and Prabhu [14]. The analysis shows that Hardik and Prabhu [14] correlation [Eq. 16–19] works well with the present data followed by Zhao et al. [12] correlation [Eq. (20)] and Cioncolini and Santini [35] correlation Eqs. (21)–(24).

Hardik and Prabhu [14] (2017)

$$\Delta P_{fric} = \phi_{LH}^2 \Delta P_L \quad (16)$$

$$\phi_{LH}^2 = 1 + \frac{C}{X_{tt}} + \frac{1}{X_{tt}^2}; \quad C = 18 \times e^{(0.14 \times d)} \quad (17)$$

$$\Delta P_L = \frac{2f_L LG^2(1-x)^2}{d \rho_L} \quad (18)$$

$$f_L = \left[Re \left(\frac{d}{D} \right)^2 \right]^{\frac{1}{20}} \frac{0.079}{Re^{0.25}}; \quad Re = \frac{Gd}{\mu_L} \quad (19)$$

Zhao et al. [12] (2003)

$$\phi_{Lo}^2 = \frac{\Delta P_{tp}}{\Delta P_{Lo}} = 1 + \left(\frac{\rho_L}{\rho_g} - 1 \right) [0.303x^{1.63}(1-x)^{0.885} Re_{Lo}^{0.282} + x^2] \quad (20)$$

Cioncolini and Santini [35] (2016)

$$\frac{dP}{dz} = -2f_h \frac{G^2}{\rho_h d} - G^2 \frac{d}{dz} \left(\frac{1}{\rho_h} \right) - \rho_h g \sin \theta \quad (21)$$

$$f_h = \left\{ 2.916 \log_{10} \left[\frac{11.78}{Re_h} + 33.57 \left(\frac{d}{D} \right)^{2.734} \right] \right\}^{-2} \quad (22)$$

$$Re_h = \frac{Gd}{\mu_h}; \quad \mu_h = \left(\frac{1-x}{\mu_L} + \frac{x}{\mu_g} \right)^{-1} \quad (23)$$

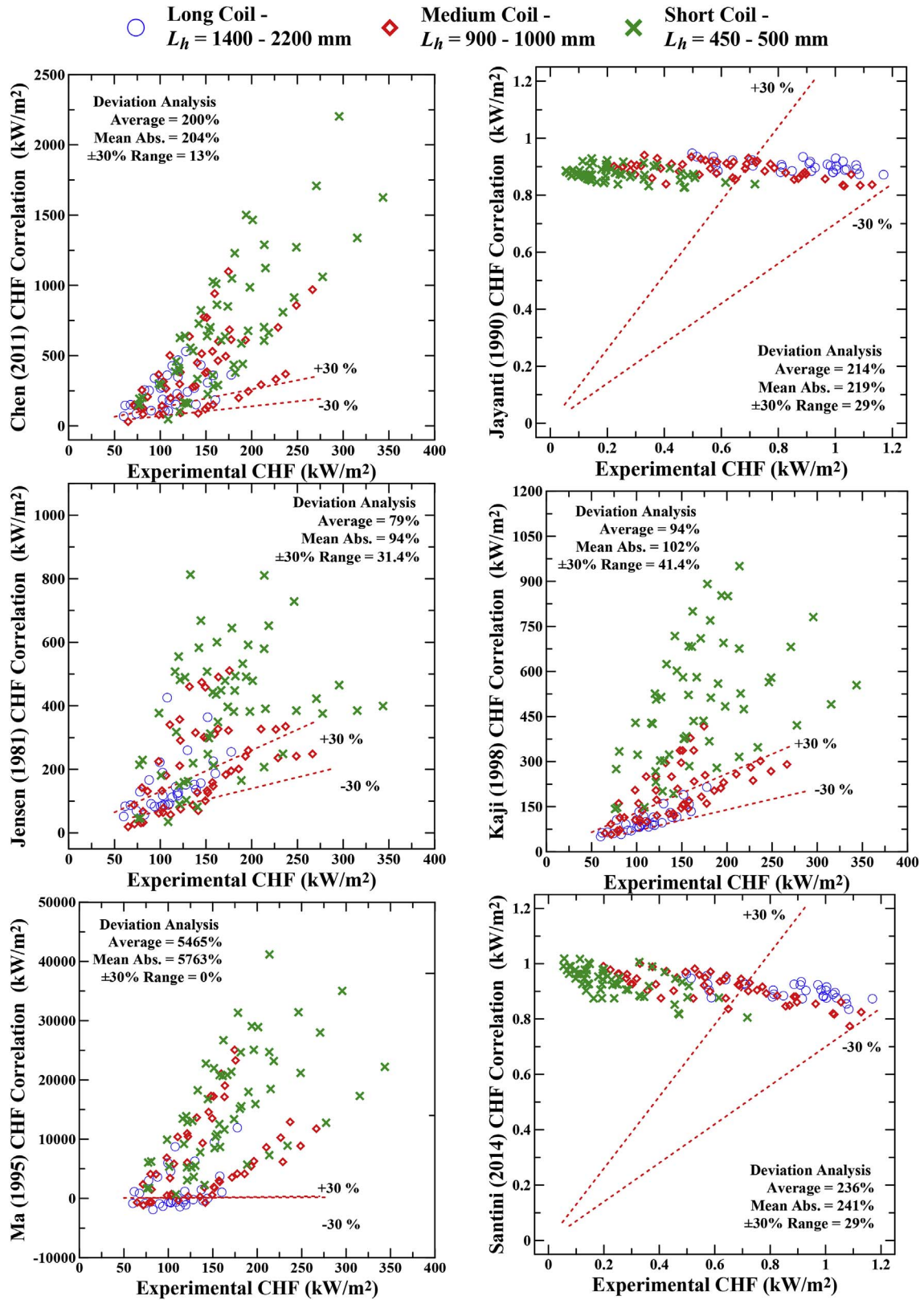


Fig. 11. Comparison of experimental CHF data with helical coil correlations.

$$\rho_h = \left(\frac{1-x}{\rho_l} + \frac{x}{\rho_g} \right)^{-1} \quad (24)$$

4.8. Comparison of critical heat flux with available helical coil correlations

The critical heat flux data are collected for all the helical coils. Method used to predict the CHF is given in the data reduction section. Length of the helical coils is changed to analyse the effect of the length

Table 5
Statistical analysis of experimental CHF data with available correlations.

Authors (Year)	Long Coils			Medium Coils			Short Coils			Overall		
	Mean Average	Absolute Mean	± 30%	Mean Average	Absolute Mean	± 30%	Mean Average	Absolute Mean	± 30%	Mean Average	Absolute Mean	± 30%
Chen et al. [39] (2011)	123	123	25	155	162	18	282	284	5.1	200	204	14.3
Jayanti and Berthoud [20] (1990)	11.1	23.0	75	80.2	85.1	34	442	442	1.7	214.2	218.6	30
Jensen and Bergles [37] (1981)	31.6	40.8	65.6	45.7	70.8	30	135	143	17.2	79.3	93.9	32.8
Kaji et al. [36] (1998)	−7.4	17.1	93.8	37.3	41.9	60	199.8	199.8	0	94.4	101.6	42.9
Ma et al. [38] (1995)	1132	2073	0	4447	4676	0	8735	8735	0	5465	5762	0
Santini et al. [24] (2014)	11.2	23.9	75	87.4	93.0	36	489	489	1.7	236	241	30.7
Kim and Mudawar [42] (2013)	−31.8	31.8	34.4	−15	24.8	62	38.8	44.8	48.3	3.4	34.7	50
Merilo [44] (1979)	−9.1	16.5	96.9	1.28	20.2	88	44.0	46.6	37.9	16.6	30.3	69.3
Shah [45] (2015)	−33	33	28.1	−24.4	26	55.1	−0.8	20.8	77.6	−16.5	25.5	58.3
Zhang et al. [43] (2006)	−10.7	16.3	100	6.7	21.9	74	44	44.8	25.9	18.2	30.1	60
Wojtan et al. [40] (2005)	−41.3	41.3	6.2	−35	35	34	−9.7	20.7	81	−25.9	30.5	47.1
Wu et al. [46] (2011)	−25.9	25.9	81.3	−18.7	18.7	94	−14.5	16.3	82.8	−18.6	19.3	86.4
Wojtan et al. [41] (2006)	−53.5	53.5	0	−37.3	37.4	36	18	31.7	53.4	−18.0	38.7	35

of test section on the CHF. The CHF data are collected for three different lengths of all helical coils. The larger length is selected based on the exit quality 1.0 during CHF. With length larger than the given length, fluid becomes saturated vapour with a quality more than 1. Helical coils are divided into three types based on heated length as short coil with heated length between 450 and 650 mm, medium coil with heated length 900–1100 mm and long coil with heated length 1400–2200 mm. Total 140 CHF data are collected for different mass flux, coil diameter, tube diameter and heated length. These data are compared with all the available correlations for CHF in helical coils. The comparison is shown in Fig. 11.

Most of the CHF correlations for helical coils are derived for long helical coil test sections. Hence, most of the correlations are matching with long coil CHF data. The correlation derived by Jayanti and Berthoud [20], Kaji et al. [36] and Santini et al. [24] are matching well with the long coil CHF data and few data of medium coil with higher exit quality. These correlations are deviating high for short helical coils CHF data. Jensen and Bergles [37] correlation is matching reasonably well with the long helical coil and a few medium helical coil CHF data. The correlation developed by Ma et al. [38] and Chen et al. [39] are not working well for the present data. The correlations of Jayanti and Berthoud [20] and Santini et al. [24] are giving almost constant value of CHF for all helical coils data. Both the correlations are giving almost same value of CHF. The statistical analysis of comparison of data with all the correlations is given in Table 5. Among all the correlations, Kaji et al. [36] correlation is working well for long and medium helical coils with lowest mean deviation.

4.9. Comparison of critical heat flux with available straight tube correlations

All the CHF correlations developed for helical coils are not able to predict the CHF in short coils. Only Kaji et al. [36] correlation works reasonably well for medium and long helical coils. Hence, in the present section, CHF data of helical coils are compared with the horizontal straight tube correlations. The comparison is shown in Fig. 12. All the straight tube correlations are predicting the data better than the helical coil correlations. This may be because the correlations of straight tube are derived using a large number of data from the literature. Helical coil correlations are derived using a limited number of data due to the lack

of CHF data in the literature. Wojtan et al. [40], Wojtan et al. [41] and Kim et al. [42] correlations have higher deviation. Zhang et al. [43] and Merilo [44] correlations are working reasonably well with data. Recent correlation developed by Shah [45] predicts the data reasonably well with a mean deviation of 25.5%. Shah [45] correlation is working well for small and medium coil. However, this correlation underpredicts the data of long coil. Zhang et al. [43] and Shah [45] correlations are difficult to use compare to Merilo [44] correlation. These correlations require an iterative method to solve. Merilo [44] correlation is easy to use and predicts the data of all the lengths of helical coils reasonably well.

Wu et al. [46] developed a simple correlation in the form of exit quality and length to tube diameter ratio. The correlation has two unknown parameters namely exit quality and heat flux. Hence, the correlation is to be solved iteratively for heat flux. If CHF is calculated by taking the exit quality from experimental data, then the correlation predicts the CHF very well with a mean deviation of 19.4%. If an iterative method is used, then the correlation is not working well. The correlation is not converging for most of the data. This may be because of very simple form of correlation which is not converging in the iteration method. The convergence of equation is highly dependent on the initial guess of exit quality. The predictions from the correlation should be taken as no more than first approximation. The comparison of Data with Wu et al. [46] correlation with exit quality taken from experimental data is shown in Fig. 13.

The statistical analysis of comparison of data with all the straight tube correlations is given in Table 5. Among all the correlations, Shah [45] correlation is working well for all the data followed by Merilo [44] correlation and Zhang et al. [43] correlation.

5. Conclusions

The experiments are performed with six helical coils having three tube diameters and two coil diameters to investigate the characteristics of the boiling heat transfer, pressure drop and CHF for refrigerant R123. Local wall temperatures in a single phase flow and in a boiling process are measured along the length of helical coil. Circumferential averaged HTC are analysed for inner side, outer side and total along the length of helical coils in a single phase flow and in a flow boiling process. Overall averaged and local heat transfer coefficients are compared with the

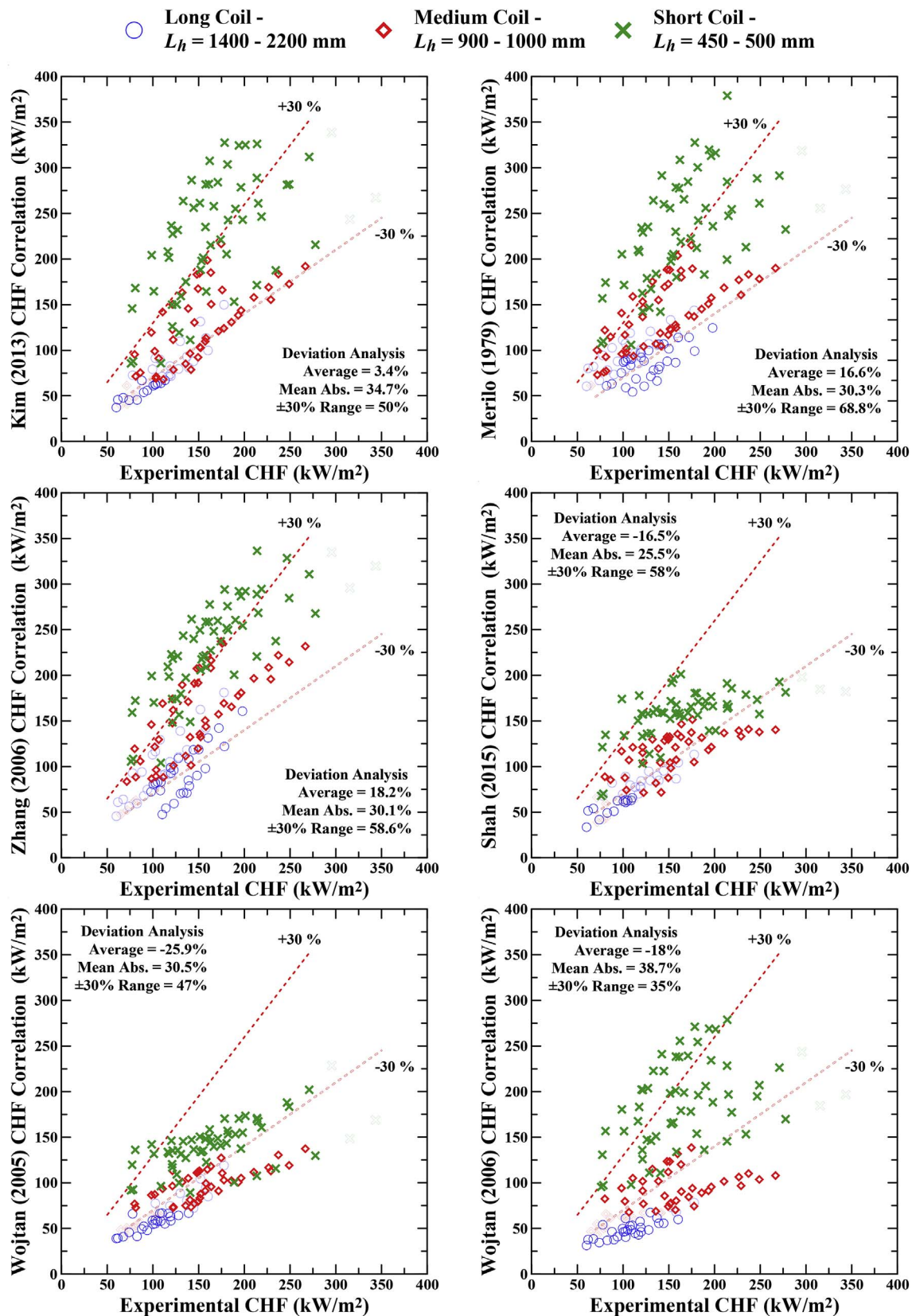


Fig. 12. Comparison of experimental CHF data with straight tube correlations.

available correlations. CHF data are collected for all the helical coils for different heated lengths. Experimental CHF data and two phase pressure drop data are compared with the available correlations. Following conclusions are drawn from the present study;

- (1) Wall temperature difference on the circumference of a helical coil is higher in the single phase flow. The difference decreases in the subcooled flow boiling. The temperature remains almost same in the nucleate boiling region. The difference in the temperature

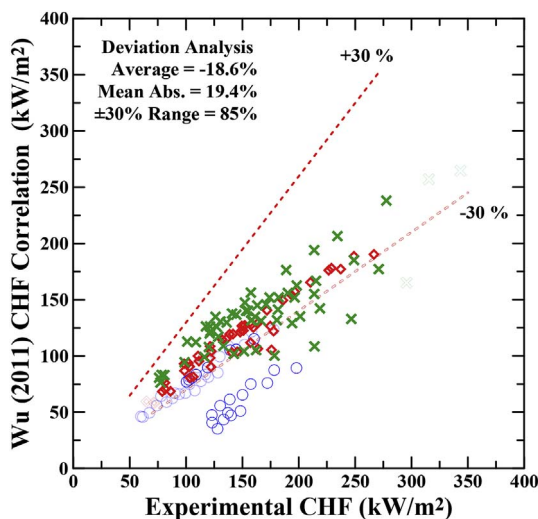


Fig. 13. Comparison of experimental CHF data with Wu et al. [46] correlation.

increases in the convective flow boiling.

- (2) In the subcooled flow boiling and in the nucleate flow boiling, the HTC across the circumference of a helical coil remains same as that of a straight tube.
- (3) In the convective flow boiling region, outer side HTC is higher than that of a straight tube. However, the inner side HTC is lower than that of a straight tube. The total circumferential averaged HTC remains same as that of a straight tube.
- (4) The inner side of a helical coil has nucleate boiling throughout the length of a coil,
- (5) The correlation developed by Hardik et al. [3] for local circumferential average inner side, outer side and total HTC in the single phase flow works well for the present study. Hence, this correlation can be used for general purpose.
- (6) Kandlikar correlation for boiling HTC works well for helical coils. Correlations other than Kandlikar underpredict the experimental data.
- (7) Two-phase pressure drop predicts well with Hardik and Prabhu [14], Zhao et al. [12] and Cioncolini and Santini [35] correlations.
- (8) Available CHF correlations for helical coils are not working well with short length helical coils. Kaji et al. [36] correlation works well for long length helical coils.
- (9) Straight tube CHF correlation works well for a helical coil. Shah [45], Zhang et al. [43] and Merilo [44] correlations work well to predict CHF in helical coils.

Contribution of the present work

The work presents an infrared thermal imaging technique to measure the local wall temperature in a helical coil. The technique helps to calculate axial and circumferential local heat transfer coefficient and to represent the actual boiling process in different portions of the helical coil. Local wall temperature and heat transfer distribution presented in the single phase and in the flow boiling process would serve as benchmark data for validating numerical results. The local heat transfer distribution aids in identifying the different regions of flow boiling. Correlations are suggested to calculate the inner side, outer side and total local HTC in a single phase flow. The work identifies the correlation which works well to predict the boiling HTC in helical coils. These correlations would be used in designing and to increase the effectiveness of heat exchangers. The data are useful in designing more compact and effective evaporators. The correlations suggested for two-phase pressure drop help to calculate the boiling pressure drop accurately. Correlation suggested for CHF would be used to design helical

coil heat transfer system for high heat transfer rate.

References

- [1] L. Santini, A. Cioncolini, M.T. Butel, M.E. Ricotti, Flow boiling heat transfer in a helically coiled steam generator for nuclear power applications, *Int J Heat Mass Transf* 92 (2016) 91–99.
- [2] H. Nariai, M. Kobayashi, T. Matsuoka, Friction pressure drop and heat transfer coefficient of two-phase flow in helically coiled tube once-through steam generator for integrated type marine water reactor, *J Nucl Sci Technol* 19 (11) (1982) 936–947.
- [3] B.K. Hardik, P.K. Baburajan, S.V. Prabhu, Local heat transfer coefficient in helical coils with single phase flow, *Int J Heat Mass Transf* 89 (2015) 522–538.
- [4] P. Naphon, S. Wongwises, A review of flow and heat transfer characteristics in curved tubes, *Renew Sustain energy Rev* 10 (5) (2006) 463–490.
- [5] S. Vashisth, V. Kumar, K.D. Nigam, A review on the potential applications of curved geometries in process industry, *Ind Eng Chem Res* 47 (10) (2008) 3291–3337.
- [6] A.M. Fsadni, J.P.M. Whitty, A review on the two-phase heat transfer characteristics in helically coiled tube heat exchangers, *Int J Heat Mass Transf* 95 (2016) 551–565.
- [7] A.M. Fsadni, J.P.M. Whitty, A review on the two-phase pressure drop characteristics in helically coiled tubes, *Appl Therm Eng* 103 (2016) 616–638.
- [8] A. Owahdi, K.J. Bell, B. Crain, Forced convection boiling inside helically-coiled tubes, *Int J Heat Mass Transf* 11 (12) (1968) 1779–1793.
- [9] Y.J. Chung, K.H. Bae, K.K. Kim, W.J. Lee, Boiling heat transfer and dryout in helically coiled tubes under different pressure conditions, *Ann Nucl Energy* 71 (2014) 298–303.
- [10] A.M. Elsayed, R.K. Al-Dadah, S. Mahmoud, A. Rezk, Investigation of flow boiling heat transfer inside small diameter helically coiled tubes, *Int J Refrig* 35 (8) (2012) 2179–2187.
- [11] K.W. Hwang, D.E. Kim, K.H. Yang, J.M. Kim, M.H. Kim, H.S. Park, Experimental study of flow boiling heat transfer and dryout characteristics at low mass flux in helically-coiled tubes, *Nucl Eng Des* 273 (2014) 529–541.
- [12] L. Zhao, L. Guo, B. Bai, Y. Hou, X. Zhang, Convective boiling heat transfer and two-phase flow characteristics inside a small horizontal helically coiled tubing once-through steam generator, *Int J Heat Mass Transf* 46 (25) (2003) 4779–4788.
- [13] S. Wongwises, M. Polsongkram, Evaporation heat transfer and pressure drop of HFC-134a in a helically coiled concentric tube-in-tube heat exchanger, *Int J Heat Mass Transf* 49 (3) (2006) 658–670.
- [14] B.K. Hardik, S.V. Prabhu, Boiling pressure drop and local heat transfer distribution of helical coils with water at low pressure, *Int J Therm Sci* 114 (2017) 44–63.
- [15] A. Cioncolini, L. Santini, M.E. Ricotti, Subcooled and saturated water flow boiling pressure drop in small diameter helical coils at low pressure, *Exp Therm Fluid Sci* 32 (6) (2008) 1301–1312.
- [16] L. Guo, Z. Feng, X. Chen, An experimental investigation of the frictional pressure drop of steam–water two-phase flow in helical coils, *Int J Heat Mass Transf* 44 (14) (2001) 2601–2610.
- [17] L. Santini, A. Cioncolini, C. Lombardi, M. Ricotti, Two-phase pressure drops in a helically coiled steam generator, *Int J Heat Mass Transf* 51 (19) (2008) 4926–4939.
- [18] H.C. Unal, M.L.G.G. Van, P.M.V. Verlaet, Dryout and two-phase flow pressure drop in sodium heated helically coiled steam generator tubes at elevated pressures, *Int J Heat Mass Transf* 24 (1981) 285–298.
- [19] S.G. Kandlikar, Development of a flow boiling map for subcooled and saturated flow boiling of different fluids inside circular tubes, *J heat Transf* 113 (1991) 190–200.
- [20] S. Jayanti, G. Berthoud, High - quality dryout in helical coils, *Nucl Eng Des* 122 (1990) 105–118.
- [21] B.K. Hardik, S.V. Prabhu, Critical heat flux in helical coils at low pressure, *Appl Therm Eng* 112 (2017) 1223–1239.
- [22] P.K. Baburajan, G.S. Bisht, S.K. Gupta, S.V. Prabhu, Measurement of subcooled boiling pressure drop and local heat transfer coefficient in horizontal tube under LPLF conditions, *Nucl Eng Des* 255 (2013) 169–179.
- [23] O.M.B. Didi, N. Kattan, J.R. Thome, Prediction of two-phase pressure gradients of refrigerants in horizontal tubes, *Int J Refrig* 25 (2002) 935–947.
- [24] L. Santini, A. Cioncolini, C. Lombardi, M. Ricotti, Dryout occurrence in a helically coiled steam generator for nuclear power application, *InEPJ Web Conf* 67 (2014) 02102 [EDP Sciences].
- [25] J.R. Moffat, Describing the uncertainties in experimental results, *Exp Therm Fluid Sci* 1 (1988) 3–17.
- [26] S.J. Kline, F.A. McClintock, Describing uncertainties in single sample experiments, *Mech Eng* (1953) 3–8.
- [27] B.K. Hardik, S.V. Prabhu, Boiling pressure drop and local heat transfer distribution of water in horizontal straight tubes at low pressure, *Int J Therm Sci* 110 (2016) 65–82.
- [28] M.M. Shah, Chart correlation for saturated boiling heat transfer: equations and further study, *ASHRAE Trans* 88 (1982) 185–196.
- [29] Z. Liu, R.H.S. Winterton, A general correlation for saturated and subcooled flow boiling in tubes and annuli, based on a nucleate pool boiling equation, *Int J heat mass Transf* 34 (1991) 2759–2766.
- [30] G.M. Lazarek, S.H. Black, Evaporative heat transfer, pressure drop and critical heat flux in a small vertical tube with R-113, *Int J Heat Mass Transf* 25 (1982) 945–960.
- [31] K.E. Gungor, R.H.S. Winterton, A general correlation for flow boiling in tubes and annuli, *Int J Heat Mass Transf* 29 (1986) 351–358.
- [32] V.E. Schrock, L.M. Grossman, Forced convection boiling in tubes, *Nucl Sci Eng* 12 (4) (1962) 474–481.
- [33] M.M. Shah, A general correlation for heat transfer during subcooled boiling in pipes and annuli, *ASHRAE Trans* 83 (1977) 202–217.

- [34] F.D. Moles, F.G. Shaw, Boiling heat transfer to subcooled liquids under the conditions of forced convection, *Trans Inst Chem Eng* 50 (1972) 76–84.
- [35] A. Cioncolini, L. Santini, Two-phase pressure drop prediction in helically coiled steam generators for nuclear power applications, *Int J Heat Mass Transf* 100 (2016) 825–834.
- [36] M. Kaji, K. Mori, T. Matsumoto, M. Oishi, T. Sawai, S. Nakanishi, Forced convective boiling heat transfer characteristics and critical heat flux in helically coiled tubes, *Nippon Kikai Gakkai Ronbunshu* 64 (1998) 3343–3349.
- [37] M.K. Jensen, A.E. Bergles, Critical heat flux in helically coiled tubes, *Trans ASME* 103 (1981) 660–666.
- [38] W. Ma, M. Zhang, X. Chen, High-quality critical heat flux in horizontally coiled tubes, *J Therm Sci* 4 (3) (1995) 205–211.
- [39] C.N. Chen, J.T. Han, T.C. Jen, L. Shao, Dry-out CHF correlation for R134a flow boiling in a horizontal helically coiled tube, *Int J Heat Mass Transf* 54 (2011) 739–745.
- [40] L. Wojtan, T. Ursenbacher, J.R. Thome, Investigation of flow boiling in horizontal tubes: Part I - a new diabatic two-phase flow pattern map, *Int J heat mass Transf* 48 (14) (2005) 2955–2969.
- [41] L. Wojtan, R. Revellin, J.R. Thome, Investigation of saturated critical heat flux in a single, uniformly heated microchannel, *Exp Therm Fluid Sci* 30 (8) (2006) 765–774.
- [42] S.M. Kim, I. Mudawar, Universal approach to predicting saturated flow boiling heat transfer in mini/micro-channels—Part I. Dryout incipience quality, *Int J Heat Mass Transf* 64 (2013) 1226–1238.
- [43] W. Zhang, T. Hibiki, K. Mishima, Y. Mi, Correlation of critical heat flux for flow boiling of water in mini-channels, *Int J Heat Mass Transf* 49 (5) (2006) 1058–1072.
- [44] M. Merilo, Fluid-to-fluid modeling and correlation of flow boiling crisis in horizontal tubes, *Int J Multiph flow* 5 (1979) 313–325.
- [45] M.M. Shah, A general correlation for critical heat flux in horizontal channels, *Int J Refrig* 59 (2015) 37–52.
- [46] Z. Wu, W. Li, S. Ye, Correlations for saturated critical heat flux in microchannels, *Int J Heat Mass Transf* 54 (1) (2011) 379–389.